



ThermaRoute AI

Thermal-Equity Copilot for Climate-Resilient Campuses

Primary SDG 11 | Secondary SDG 3 and SDG 13

AI + RAG

risk scoring, guidance retrieval, action planning

Rare idea

hyperlocal heat exposure, not generic weather alerts

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Heat risk is hyperlocal, but most tools are city-level.

- The same campus can contain cool shaded paths and dangerous heat pockets.
- People who walk, wait, clean, guard, deliver, or maintain facilities cannot always avoid peak heat.
- Weather alerts rarely explain where to add shade, water, rest breaks, or safer routing.

Common project blind spot

Many sustainability projects stop at broad prediction. ThermaRoute AI turns climate risk into micro-location decisions.



Extreme heat is a health, work, and city-resilience problem.

489k

approx. annual heat-related deaths globally in 2000-2019 studies

2.4B

workers estimated by ILO as exposed to excessive heat

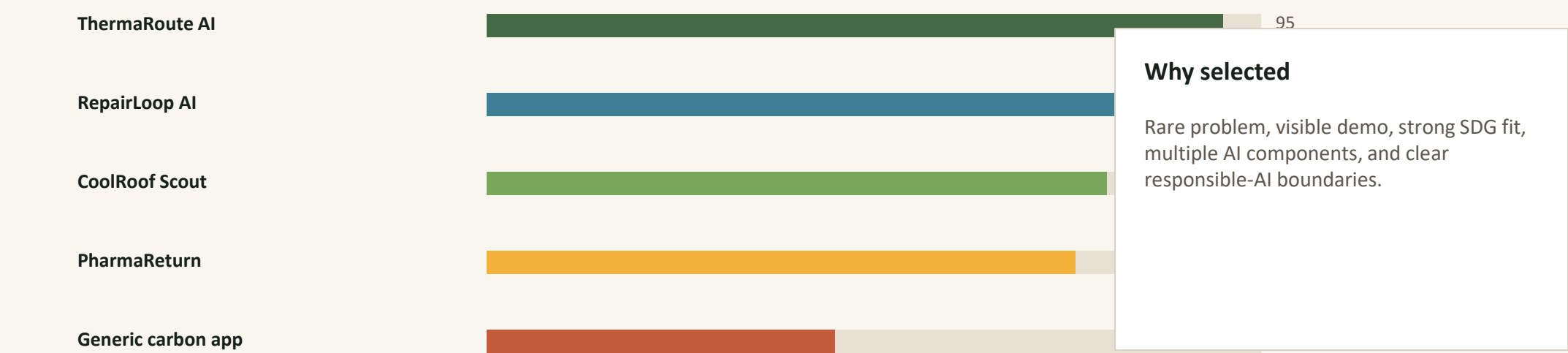
UHI

urban heat islands make cities warmer than surroundings

The sustainability challenge is not only rising temperature. It is unequal exposure: who has shade, water, schedule control, and safer routes.

Sources: WHO Heat and health fact sheet; ILO Heat at work; IPCC AR6 Urban Areas Fact Sheet.

The chosen project beats common ideas on originality and demo value.





How might we use AI to predict hyperlocal heat exposure and recommend safer routes, schedules, rest breaks, water access, and shade interventions so that campuses and outdoor work zones become more climate-resilient, inclusive, and sustainable?



SDG 11 is the anchor: safer and more resilient local environments.

Primary SDG 11

Sustainable Cities and Communities: identify heat hotspots and prioritize shade, water, cool routes, and local adaptation.

Secondary SDG 3

Good Health and Well-being: prevent heat stress through practical risk guidance and human escalation.

Secondary SDG 13

Climate Action: prepare institutions for more frequent and intense extreme heat conditions.

ThermaRoute AI protects people with unequal exposure and low schedule control.

Students

long walks across exposed routes

Security staff

fixed outdoor posts

Sanitation teams

physical work in open areas

Maintenance workers

rooftops and service yards

Administrators

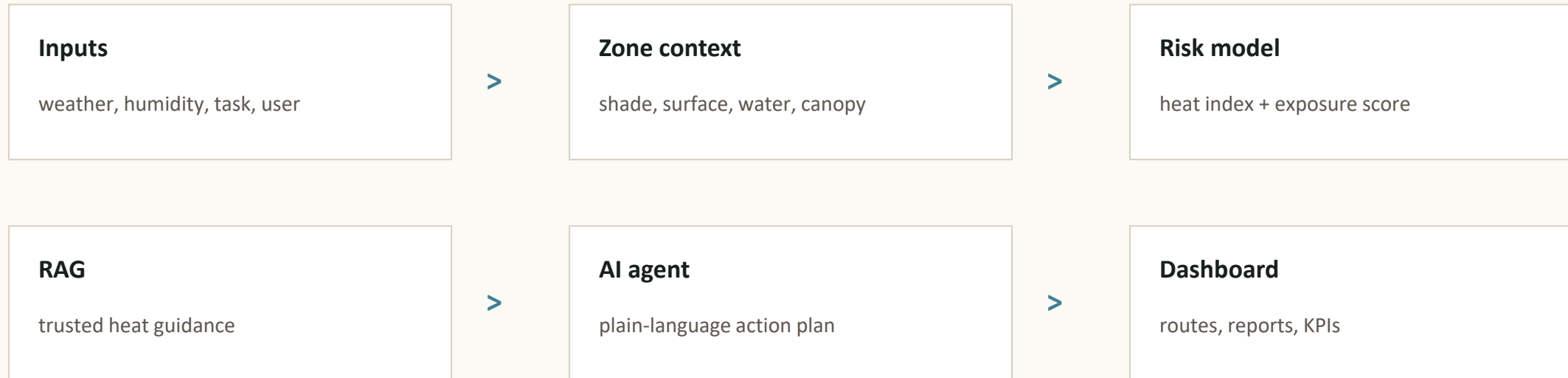
heat action and infrastructure planning

A decision-support loop, not just another chatbot.



- Risk score combines weather, zone, task, user group, duration, and timing.
- RAG adds trusted guidance before the AI writes an action plan.
- Community reports convert lived experience into adaptation priorities.

The architecture combines scoring, RAG, planning, and analytics.



The prototype uses mock campus data but production-ready data fields.

surface type

shade score

water points

tree canopy

temperature modifier

people exposed

task duration

user group

Future inputs: live weather API, IoT sensors, GPS routes, GIS layers, image-based shade detection.

Maintenance worker on a hostel rooftop at midday

Weather

38 C, 61% humidity

Task

heavy work, 45 minutes

Place

metal roof, shade 10/100

Access

no water point

Signal

dizziness reported

Expected result: Extreme risk. Postpone non-essential work, add shade and water, and escalate symptoms to human support.

Four dashboard tabs make the demo easy to judge.

Assess

Run a task scenario and generate score, tier, rest plan, hydration plan, recommendations, and evidence.

Zones

View campus hotspots and safer route options based on shade and water access.

Reports

Submit anonymous heat observations to build a community sensing loop.

Sources

Show the RAG knowledge base and trusted evidence used by the AI assistant.

The system is designed to assist, explain, and escalate.

Fairness	includes workers, students, staff, and elderly visitors
Transparency	shows score, tier, reasons, and sources
Privacy	anonymous reports only
Safety	does not diagnose; symptoms route to humans
Hallucination control	RAG plus deterministic fallback

Expected impact spans environment, people, and operations.

Environmental

More shade, trees, cool surfaces, and climate-resilient campus planning.

Social

Safer schedules and routes for people with unequal heat exposure.

Economic

Less disruption, smarter maintenance timing, and better targeting of low-cost interventions.

The project can be evaluated with measurable sustainability KPIs.

- Zones assessed and high-risk zones improved.
- Peak-hour outdoor tasks reduced or rescheduled.
- Water and shade access coverage increased.
- Anonymous heat reports resolved by administrators.
- Safer route recommendations used during heat events.



The demo is complete, local, and upgrade-ready.

Frontend

React, Vite, Tailwind CSS, responsive dashboard, lucide icons.

Backend

FastAPI, Pydantic, SQLite, heat-risk service, RAG service, AI adapter.

AI integration

Works offline with deterministic mock AI; can connect to IBM Granite through an OpenAI-compatible endpoint.

Designed for student feasibility today and real campus pilots tomorrow.

From campus prototype to city heat-action platform



ThermaRoute AI turns invisible heat exposure into visible, responsible, and actionable climate adaptation.

References and evidence base

- WHO: Heat and health fact sheet.
- ILO: Heat at work - implications for safety and health.
- IPCC AR6: Urban Areas Fact Sheet and urban heat island evidence.
- NDMA India: Heat Wave guidance and thresholds.
- PIB / IMD-linked public heatwave safety guidance.

All claims in the demo are planning guidance, not medical diagnosis. Human escalation is required when symptoms or emergency conditions appear.